

✅ LIVE — MAINNET RUNNING & MINING ACTIVE

# TARCOIN

## [ TAR ]

A UTXO-Based Decentralized Cryptocurrency Ecosystem
Built on Battle-Tested Proof-of-Work Architecture

| VERSION | PUBLISHED | ALGORITHM | LICENSE |
|---|---|---|---|
| 1.0.0 | June 2026 | SHA256d PoW | MIT |

| 50B | 50,000 | 10 min | 400K |
|---|---|---|---|
| TOTAL SUPPLY (TAR) | BLOCK REWARD (TAR) | BLOCK TIME | HALVING INTERVAL |

| P2P:19333 |
|---|
| MAINNET PORT |

# Table of Contents


00 Abstract

01 Introduction

02 Technical Architecture

03 Network Architecture

04 Consensus Rules

05 Ecosystem Treasury Security

06 Security Model

07 Infrastructure Architecture

08 Deployment Architecture

09 Testing & Audit

10 ✅ Launch Checklist

11 Governance

12 Conclusion & References



ABSTRACT

TARCOIN (TAR) is a fully decentralized, UTXO-based cryptocurrency built on a battle-tested Proof-of-Work blockchain foundation derived from Bitcoin Core. It inherits the complete security model — SHA256d Proof-of-Work, UTXO transaction model, secp256k1 elliptic curve cryptography, ECDSA signatures, Bitcoin Script, Merkle trees, difficulty retargeting, SegWit, and longest-chain consensus — while introducing a distinct identity, custom supply parameters, and a complete web/mobile ecosystem.

Unlike experimental blockchain projects that modify consensus rules or introduce centralized components, TARCOIN remains faithful to the proven principles of decentralized consensus, ensuring maximum security, ASIC miner compatibility, and long-term stability.

SECTION 01

# Introduction

## 1.1 Philosophy

TARCOIN follows the original Bitcoin philosophy: a peer-to-peer electronic cash system that is decentralized, permissionless, and immutable. Every design decision prioritizes **security over convenience**, **decentralization over speed**, and **immutability over flexibility**.

## 1.2 Design Goals

1. **Battle-Tested Security** — Preserve all Bitcoin consensus rules and cryptographic primitives
2. **ASIC Mining Compatibility** — SHA256d algorithm supports existing Bitcoin mining hardware
3. **Immutable Supply** — Fixed 50 billion TAR supply, enforced by consensus
4. **Decentralized Operation** — No admin controls, no backdoors, no central authority
5. **Privacy-Preserving Development** — Secure, anonymous development and deployment
6. **Production Infrastructure** — Explorer, API, mining pool, wallets, monitoring

SECTION 02

# Technical Architecture

## 2.1 Blockchain Foundation

TARCOIN is derived from Bitcoin Core v31.x. Only the following components are modified:

- **Supply constants** (`amount.h`)
- **Difficulty adjustment** (`pow.cpp` — Dark Gravity Wave v3 per-block retargeting and 6-hour Emergency Difficulty Adjustment)
- **Network parameters** (`chainparams.cpp` — magic bytes, ports, DNS seeds, checkpoints)
- **Address prefixes** (bech32: `tar1`, base58: `T`)
- **Branding** (coin name, ticker, UI identity)
- **Genesis block** (unique hash, treasury allocation)

Everything else — validation logic, consensus engine, networking, mempool, wallet encryption, RPC interface — remains **identical to upstream Bitcoin Core**, ensuring full compatibility with the Bitcoin security model.

## 2.2 Cryptographic Primitives

| COMPONENT | STANDARD |
|---|---|
| Hashing Algorithm | SHA256d (double SHA-256) |
| Elliptic Curve | secp256k1 |
| Signature Algorithm | ECDSA |
| Address Format | Bech32 (SegWit) and Base58 (Legacy) |
| Merkle Trees | Binary Merkle (Bitcoin standard) |
| Script Language | Bitcoin Script |

## 2.3 Supply Schedule

TOTAL SUPPLY

**50,000,000,000 TAR**

ECOSYSTEM TREASURY (20%)

**10,000,000,000 TAR**

PUBLIC MINING SUPPLY (80%)

**40,000,000,000 TAR**

BLOCK REWARD (ERA 1)

**50,000 TAR**

## Emission Schedule

```
Era 1:  Block       0 - 400,000  |  50,000 TAR/block  |  ~7.6 yrs  |  20,000,000,000 TAR
Era 2:  Block 400,001 - 800,000  |  25,000 TAR/block  |  ~7.6 yrs  |  10,000,000,000 TAR
Era 3:  Block 800,001 - 1,200,000|  12,500 TAR/block  |  ~7.6 yrs  |   5,000,000,000 TAR
Era 4:  Block 1,200,001-1,600,000|   6,250 TAR/block  |  ~7.6 yrs  |   2,500,000,000 TAR
...emission continues halving every 400,000 blocks until max supply is reached
```

SECTION 03

# Network Architecture

## 3.1 Network Parameters

| PARAMETER | MAINNET | TESTNET | REGTEST |
|---|---|---|---|
| P2P Port | 19333 | 29333 | 18444 |
| RPC Port | 19332 | 29332 | 18443 |
| Magic Bytes | 0xfabfb5da | 0x0b110907 | 0xfabfb5da |
| Address Prefix | tar1 / T | tar1 / T | tar1 / T |

## 3.2 Mainnet Parameters

| PARAMETER | VALUE |
|---|---|
| Coin Name | TARCOIN |
| Symbol | TAR |
| Consensus | SHA256d Proof-of-Work |
| Block Time | 10 minutes (Bitcoin-standard) |
| Initial Block Reward | 50,000 TAR |
| Difficulty Retarget | Every block (Dark Gravity Wave v3, 24-block rolling) |
| Emergency Adjustment (EDA) | 50% difficulty cut if block time > 6 hours |
| Halving Interval | 400,000 blocks |
| Maximum Supply | 50,000,000,000 TAR |
| Mineable Supply | 40,000,000,000 TAR (80%) |
| Ecosystem Treasury | 10,000,000,000 TAR (20%) |
| Smallest Unit | Stella / sTAR (1 TAR = 100,000,000 sTAR) |
| P2P Port | 19333 |
| RPC Port | 19332 |
| Bech32 Address Prefix | tar1 |

### 3.3 DNS Seed Nodes

Seed nodes are deployed across multiple cloud providers (DigitalOcean, AWS, Hetzner, Vultr) to ensure network resilience. Each seed node runs a full archive node.

### 3.4 Peer-to-Peer Protocol

TARCOIN uses the standard Bitcoin-compatible P2P protocol:

- `Version/VerAck` — initial handshake
- `Inv/GetData` — inventory propagation
- `Headers/GetHeaders` — block header sync
- `Block/Tx` — data propagation
- `Addr` — peer discovery
- `Ping/Pong` — keepalive
- `FeeFilter` — fee relay

SECTION 04

# Consensus Rules

## 4.1 Proof-of-Work & Difficulty Adjustment

TARCOIN employs the SHA256d (double SHA-256) Proof-of-Work algorithm, ensuring full hardware compatibility with all existing SHA-256 ASIC miners.

Unlike Bitcoin's static 2016-block retarget cycle, TARCOIN implements **Dark Gravity Wave v3 (DGW v3)** for dynamic, continuous difficulty adjustment:

- **Per-Block Retargeting:** Network difficulty recalculates dynamically on **every single block** based on a rolling window of the preceding 24 blocks. This prevents multi-pool exploitation, eliminates "hashrate hopping," and ensures steady block production regardless of sudden hashrate fluctuations.
- **Emergency Difficulty Adjustment (EDA):** If network hashrate drops drastically and no block is mined for more than **6 hours**, consensus rules trigger an automatic difficulty reduction (cutting difficulty by 50% / doubling target: `bnTarget *= 2`). This guarantees the blockchain never stalls under low hashrate conditions.
- **Target Block Time:** 10-minute average block time maintained smoothly across all network hashrate scales.

## 4.2 Block Validation

Block validation follows the Bitcoin Core consensus standard:

- Verify block header hash meets difficulty target
- Verify Merkle root matches transactions
- Verify all transaction inputs reference valid UTXOs
- Verify no double-spends within the block
- Verify block size ≤ 1MB (base) / 4MB (SegWit)
- Verify signature operations limits
- Verify coinbase maturity

## 4.3 Genesis Block

The TARCOIN genesis block was successfully mined and verified on mainnet.

| | |
|---|---|
| Block Hash | 0000e37ee7aa8a88d1254ee3fe7c497c8fdaff36b29747eb64d8da68fbd9939e |
| Merkle Root | eaf980618b3cde94360d83c4937c3bcebcba7cc1db9f3276eb8b180da3a2e921 |
| Timestamp | 1782181262 (2026-06-23 02:21:02 UTC) |
| nBits | 0x1f00ffff |
| Nonce | 65067 |
| Block Reward | 50,000 TAR |
| PoW Target | 0000ffff0000000000000000000000000000000000000000000000000000 |
| powLimit | 0000ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff |
| Height | 0 |

The genesis block is **hardcoded** into `chainparams.cpp` and verified by `assert()` on every node startup.

SECTION 05

# Ecosystem Treasury Security

## 5.1 Offline Generation

The ecosystem treasury wallet is created entirely offline using an **air-gapped machine** that has never been connected to the internet.

1. Generate wallet on air-gapped machine
2. Record public address only (publicly verifiable)
3. Create multiple encrypted backups (USB, paper, steel)
4. Transfer public address to development environment
5. Hardcode address into `chainparams.cpp`
6. Destroy intermediate systems

## 5.2 Private Key Security

⚡

🚫 **MUST NEVER** touch any internet-connected system

🚫 **MUST NEVER** be stored in repositories or cloud systems

🚫 **MUST NEVER** be logged or screenshotted

✅ **MUST** have geographically separated backups

✅ **MUST** follow industry-standard cold storage best practices

## 5.3 Transparency & Management

- All reserve transactions are publicly recorded on the blockchain
- Reserve deployments will be communicated through official channels
- The blockchain consensus layer operates independently of any developer or entity
- Mining rewards are fully decentralized with no central authority over block production

SECTION 06

# Security Model

## 6.1 Attack Mitigation

### 51% Attack Protection (Early Network)

- Encourage distributed mining participation
- Deploy multiple mining pools
- Public mining awareness campaign
- Optional temporary checkpoints if required (for initial security only)
- Rapid hashrate growth through ASIC compatibility

### Reorg Protection

- Bitcoin-standard reorganization handling
- 100-block maturity for coinbase outputs
- Merkle tree verification
- Checkpoints at regular intervals

### Double-Spend Protection

- Full node validation
- Mempool orphan detection
- Replace-by-fee (RBF) support
- UTXO set verification

## 6.2 Wallet Security

- AES-256 encrypted wallet.dat files
- BIP39/BIP32 mnemonic seed phrases
- Passphrase protection
- Offline transaction signing (cold storage)
- Hardware wallet compatibility roadmap

## 6.3 Network Security

- Full node validation only (no SPV reliance)
- Tor/I2P support for privacy
- Encrypted P2P connections (BIP324)

- DoS protection (limit counters, penalty system)

SECTION 07

# Infrastructure Architecture

## 7.1 Explorer

The blockchain explorer provides real-time block, transaction, and address lookup:

- 📦 Real-time block lookup
- 💸 Transaction explorer
- 📊 Difficulty & hashrate charts
- 🏦 Supply tracking & rich list
- 🔴 WebSocket live updates
- ⚡ Redis-cached API

## 7.2 REST API

RESTful API with comprehensive blockchain data endpoints, Swagger documentation, rate limiting, and DDoS protection.

## 7.3 Mining Pool

Full-featured mining pool on Stratum port 3333 with worker management, share tracking via Redis, payout engine, and ASIC miner compatibility.

## 7.4 Website

Built with Next.js featuring a cyberpunk/gold-black aesthetic, Framer Motion animations, Three.js particle effects, responsive design, and SEO optimization.

SECTION 08

# Deployment Architecture

## 8.1 Docker Stack

All services are containerized for deterministic deployment.

| SERVICE | DESCRIPTION |
|---|---|
| tarcoind | Blockchain full node |
| explorer | Blockchain explorer frontend + backend |
| api | REST API server |
| mining-pool | Mining pool + Stratum server |
| website | Next.js public website |
| nginx | Reverse proxy (SSL termination, DDoS protection) |
| redis | Cache layer |
| postgres | Relational database |
| prometheus / grafana / loki | Monitoring & observability stack |

## 8.2 Infrastructure Requirements

| SERVICE | CPU | RAM | STORAGE | NETWORK |
|---|---|---|---|---|
| tarcoind (archive) | 8 cores | 16 GB | 1+ TB SSD | 1 Gbps |
| tarcoind (pruned) | 4 cores | 8 GB | 50 GB SSD | 100 Mbps |
| Explorer + API | 4 cores | 8 GB | 100 GB SSD | 1 Gbps |
| Mining Pool | 4 cores | 8 GB | 50 GB SSD | 1 Gbps |
| Website | 2 cores | 4 GB | 20 GB SSD | 100 Mbps |
| Monitoring | 2 cores | 4 GB | 100 GB SSD | 100 Mbps |

### 8.3 Cloud Providers Supported

- DigitalOcean
- AWS (EC2, EKS)
- Hetzner
- Vultr

SECTION 09

# Testing & Audit

## 9.1 Mandatory Audits

| # | AUDIT TYPE | SCOPE |
|---|---|---|
| 1 | Consensus Audit | Verify all consensus rules match Bitcoin Core |
| 2 | Supply Audit | Verify supply constants, halving schedule, treasury allocation |
| 3 | Wallet Penetration Test | Secure key storage, encryption, signing |
| 4 | RPC Security Audit | No unauthorized access, injection, or exposure |
| 5 | API Security Testing | Rate limiting, SQL injection, XSS, CSRF |
| 6 | Memory Analysis | No leaks in long-running processes |
| 7 | Reorg Attack Simulation | Network partition handling |
| 8 | 51% Attack Simulation | Chain reorganization resilience |
| 9 | Mining Stress Test | Pool and Stratum server capacity |
| 10 | P2P Network Test | Peer discovery, propagation, DoS resilience |

## 9.2 Deterministic Builds

All releases must be:

- Deterministic (same source → same binary)
- Reproducible (anyone can verify)
- Publicly auditable
- Signed with release keys

SECTION 10

# Launch Checklist

## PRE-LAUNCH

✅ 18 / 18 COMPLETE

- ✓ Genesis block mined and verified ( `0000e37e...fbd9939e` )
- ✓ TARCOIN Core v31.x compiled and running
- ✓ Mainnet node boots and passes PoW validation
- ✓ `powLimit` aligned with genesis `nBits` ( `0x1f00ffff` )
- ✓ Genesis hash and merkle root `assert()` verified on startup
- ✓ `CLIENT_VERSION_IS_RELEASE = true` set
- ✓ Seed nodes deployed (DNS seeds live)
- ✓ Explorer backend verified
- ✓ Wallet binaries built and signed
- ✓ Mining pool configured and tested
- ✓ Infrastructure stress-tested
- ✓ Monitoring stack operational
- ✓ DNS records configured
- ✓ SSL certificates issued
- ✓ Cloudflare configured
- ✓ CI/CD pipeline validated
- ✓ Deterministic builds confirmed
- ✓ Source code publicly released on GitHub

## LAUNCH

✅ 8 / 8 COMPLETE

- ✓ Seed nodes go online

- ✓ Explorer goes online
- ✓ Mining pool activates
- ✓ Wallet downloads available
- ✓ Website published
- ✓ API operational
- ✓ Social media announcements
- ✓ Mining campaigns begin

## POST-LAUNCH

4 / 4 COMPLETE

- ✓ Network monitoring active
- ✓ Community support channels live
- ✓ Bug bounty program launched
- ✓ Ongoing security audits in progress

**Transparency Statement:** All critical network parameters, supply allocations, genesis data, and consensus rules are publicly documented and independently verifiable from the TARCOIN source code.

## SECTION 11 Governance

TARCOIN operates without any centralized governance authority. The blockchain is governed solely by its immutable consensus rules, enforced by network participants running full nodes.

- 🔒 No foundation, corporation, or legal entity controls the network
- 🔒 No administrative keys or backdoors exist
- 🔒 No single entity has authority to upgrade or modify the protocol
- 🔒 No governance tokens or on-chain voting mechanisms
- 🌐 Development is community-driven through open-source contributions
- 🔒 No individual can alter supply schedule, consensus rules, or treasury after genesis

The ecosystem treasury of 10 billion TAR is allocated at genesis to a publicly verifiable cold storage address. The treasury allocation is permanently recorded in the genesis block and remains publicly auditable on-chain.

## SECTION 12 Conclusion

⚡

*TARCOIN represents a return to the foundational principles of decentralized cryptocurrency — a secure, permissionless, UTXO-based blockchain network that anyone can mine, use, or build upon. By preserving a battle-tested Proof-of-Work codebase while introducing a unique identity, custom supply parameters, and a complete ecosystem, TARCOIN provides a reliable foundation for long-term value storage and peer-to-peer transactions.*

**TARCOIN is not an experiment.**

It is battle-tested Proof-of-Work architecture, reborn for a new era.

## References


1. Nakamoto, S. "Bitcoin: A Peer-to-Peer Electronic Cash System" (2008)
2. Bitcoin Core — https://github.com/bitcoin/bitcoin
3. BIP 32 — Hierarchical Deterministic Wallets
4. BIP 39 — Mnemonic Code for Generating Deterministic Keys
5. BIP 141 — Segregated Witness (SegWit)
6. BIP 173 — Bech32 Address Format
7. Dark Gravity Wave v3 (DGW v3) — Continuous Difficulty Adjustment Algorithm





*Released under the MIT License.*
*For informational purposes only. Not financial advice.*